

ENGR 102 Summer Bridge

Day 12 Student Quiz

For Loops, Range, Counters, and Accumulators

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace the complete loop state.

5 points

```
total = 1
count = 0
for value in range(3, 10, 2):
    total = total * 2 + value
    if total % 3 == 0:
        count = count + 1
print(total, count)
```

Show value, total, and count after every pass, then give exact output.

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Question 2. Diagnose and repair an inclusive-bound bug.

3 points

```
product = 1
for factor in range(2, 5):
    product = product * factor
print(product)
```

The intent is $2 * 3 * 4 * 5$. Give actual, expected, cause, repair, and one test.

Question 3. Program construction

12 points

Write one complete Python program that satisfies every requirement.

- Read days as a positive integer and target as a float.
- If days is not positive, print invalid.
- Otherwise read exactly days floating-point usage values, compute their total, and count values strictly above target.
- Print total and above_count on separate lines. Include one boundary test for usage equal to target.

[illegible]

Test evidence / expected result